

DEPARTMENT OF MATHEMATICS, RIT - BANGALORE 54

QUIZ TEST

DMS

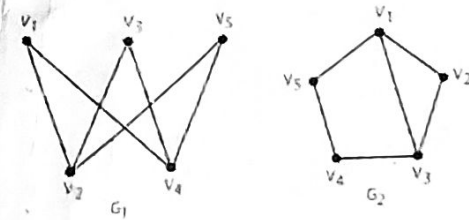
22.02.2024

4.45pm TO 5.15pm

ANSWER ALL QUESTIONS - EACH CARRIES 2 MARKS

1. Without using truth tables verify $((P \wedge \neg Q) \vee (P \wedge Q))$ is logically equivalent to $(P \wedge Q)$.
2. Draw the Hasse diagram of the poset $(\{2,4,5,10,12,20,25\}, /)$. Find the maximal and minimal elements?
3. Find an explicit formula for $a_n = 2a_{n-1} + 1$ with $a_1 = 7$.

4. Verify the following graphs are isomorphic or not? Justify your answer.



5. Find the minimal spanning tree by Kruskal's algorithm for the following graph

